

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
CO5_AT1 — UNCERTAINTY VISUALIZATION LAB

Course Code:	DSA06	Course Title:	Data Handling and Visualization
CO Assessed:	CO5 – Analysis (BL4)	Assessment:	Assessment Tool 1 – Uncertainty Visualization Lab
Weightage:	20% of CIA	Total Marks:	25 (5 Questions × 5 Marks)
CO5 Statement:	<i>Analyze and construct visualizations that communicate uncertainty using frequency framing, confidence intervals, trend uncertainty bands, HOPs, bootstrapping, and appropriate axis scales.(BL4)</i>		
Student Name:	Nivya Sri T	Reg. No.:	192424131
Section / Batch:	2024	Date:	27-08-26

Code of Conduct

I Nivya Sri T (192424131) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Nivya Sri T

Instructions

- This test contains 5 questions, each carrying 5 marks. Total: 25 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 90 minutes.
- Show all supporting calculations (e.g., confidence interval bounds, residual standard deviation, bootstrap resample means) where relevant.
- Every diagram must include a title, correctly labeled axes, and a legend where relevant.
- Diagrams may be hand-drawn or clearly described in words if drawing tools are unavailable.

Annexure A – UNCERTAINTY VISUALIZATION LAB

Rubrics for Calculation ASSESSMENT RUBRIC
AT1 — Uncertainty Visualization Lab
CO5 · BT4: Analyze ·

This rubric is used to assess student responses to the CO5-AT1 Uncertainty Visualization Lab problems, in which students construct and interpret uncertainty visualizations (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, and axis scale choice) from given data.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25%	Demonstrates thorough understanding of the relevant uncertainty-visualization technique (frequency framing, confidence intervals, trend bands, HOPs, bootstrapping, or axis scale choice) and correctly applies it to the given data.	Good understanding of the relevant technique, with only minor gaps in application.	Basic understanding, but with some misconceptions about the correct technique or how to apply it.	Poor understanding of core concepts; applies a technique fundamentally unsuited to the data.
Explanation	25%	Provides detailed, well-reasoned analysis explicitly tied to the specific values in the dataset, including all required calculations.	Provides clear analysis with adequate reasoning tied to the data.	Analysis is present but generic or only loosely connected to the specific dataset given.	Little or no analytical reasoning provided; answers restate the data without interpretation.
Application	20%	Effectively applies construction and calculation techniques to produce a complete, accurate uncertainty visualization correctly matched to the data.	Applies techniques to produce a correct visualization, with only minor errors.	Limited application; the visualization partially represents the uncertainty but has notable errors.	Fails to apply appropriate techniques; the visualization does not correctly represent the uncertainty.
Organization	15%	The response is clearly structured, with construction and analysis presented in a logical sequence	The response is organized, with only minor issues in flow or clarity.	Basic structure; construction and analysis are present but the response lacks clarity or	Poorly organized; the response is incoherent or key parts are left unaddressed.

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
		with coherent overall flow.		sequence.	
Accuracy	10%	All parts of the answer — construction, calculations, and analysis — are completely accurate and well-suited to the given data.	Mostly accurate, with only minor omissions or a small error.	Partially correct; some elements are missing, mismatched to the data, or incorrect.	Largely incorrect or inappropriate, with major gaps across multiple parts.

Scoring Guide

For each criterion, award a performance level (1–4) based on the descriptors above. Multiply the level achieved (as a percentage of 4, i.e., Level 4 = 100%, Level 3 = 75%, Level 2 = 50%, Level 1 = 25%) by the criterion's weightage, then sum across all five criteria to obtain the total score out of 100%.

Faculty In-charge	Course Coordinator	Head of Department
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

PROBLEM 36: Visualizations Along Linear and Log Axes

1. Ratios Between Consecutive Values

- $1200 / 50 = 24\times$
- $40,000 / 1,200 \approx 33\times$
- $900,000 / 40,000 = 22.5\times$

The data grows multiplicatively, not additively.

2. Linear Scale

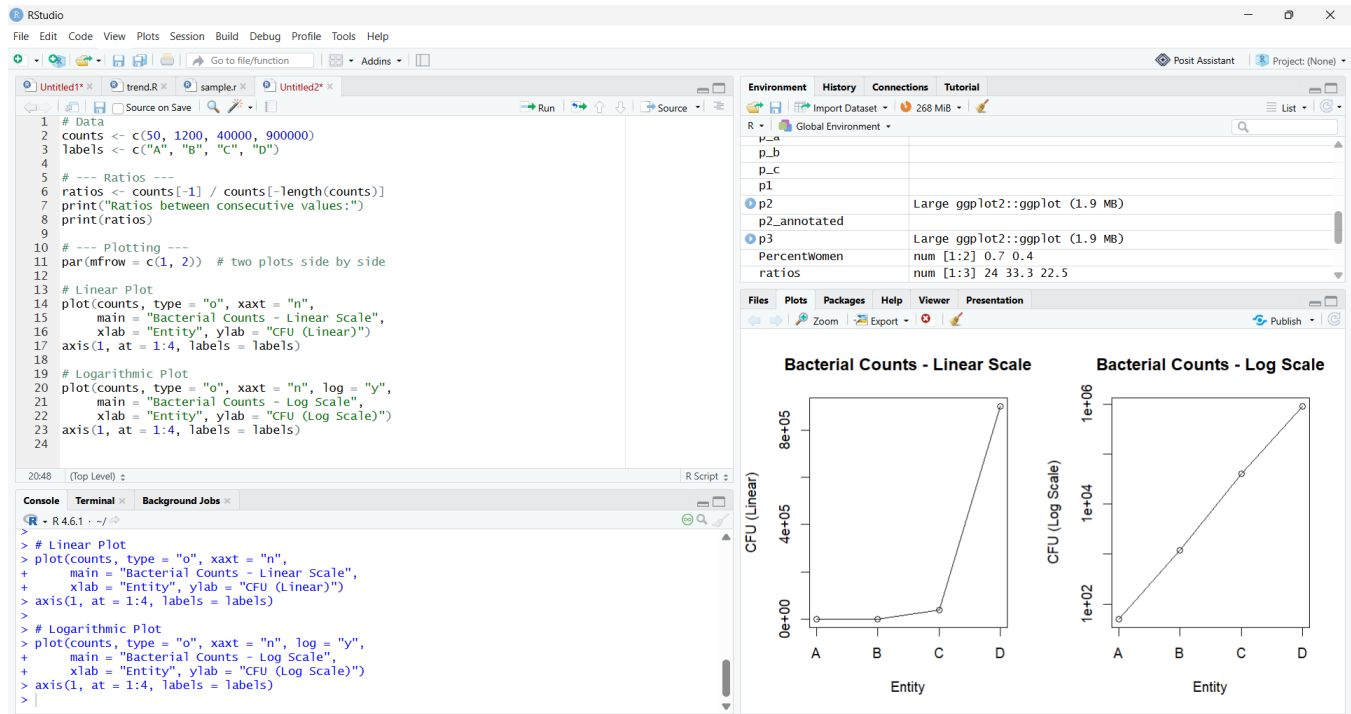
- Smaller values (50, 1200) appear almost flat.
- Larger values dominate the plot.
- The growth pattern is not visible.

3. Logarithmic Scale

- Log scale spreads values according to powers of 10.
- Multiplicative jumps (22–33 $\times$) become evenly spaced.
- The exponential growth pattern becomes clear and readable.

4. Conclusion

Logarithmic axis shows the structure more clearly because bacterial counts increase by large multiplicative factors, and log scale converts multiplicative growth into a smooth, visible pattern.



PROBLEM 37: Framing Probabilities as Frequencies

Natural Frequency Statement

A 22% probability can be reframed as:

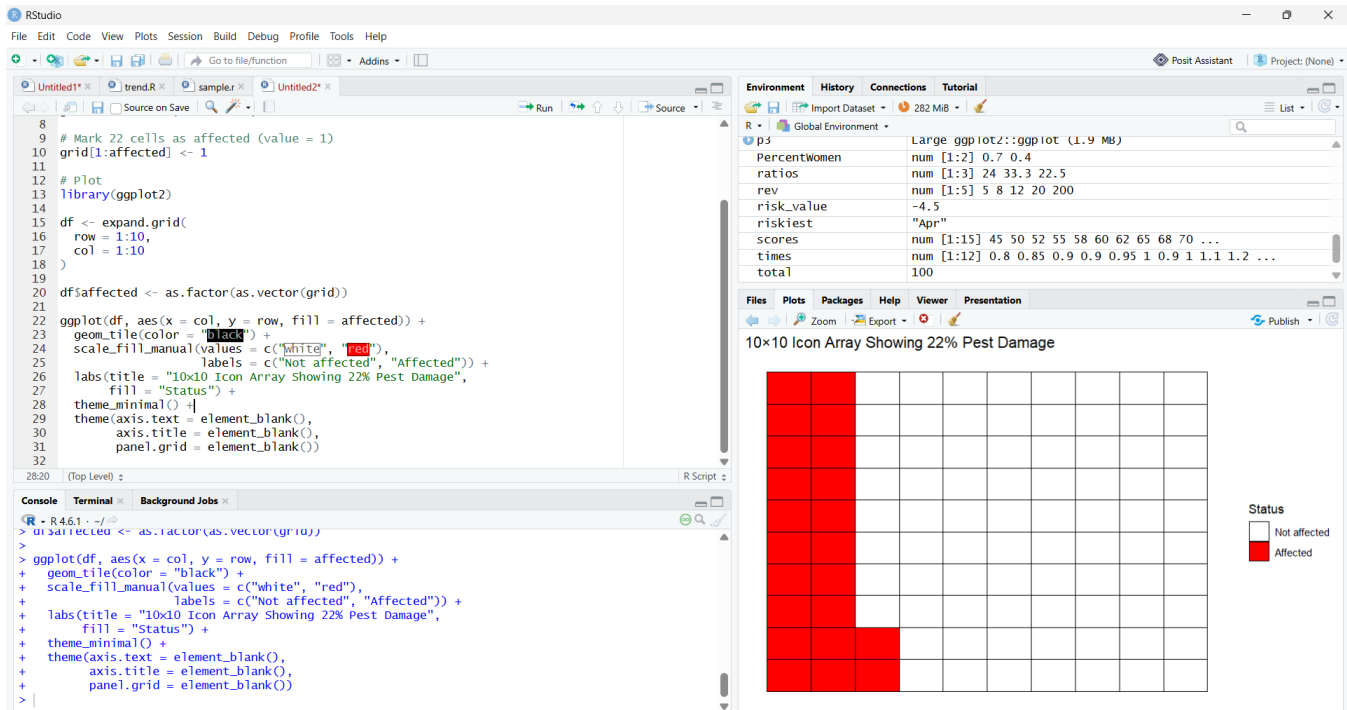
Out of 100 crop fields, about 22 would be affected by pest damage.

This makes the probability easier to understand because it uses actual counts instead of percentages.

10 $\times$ 10 Icon Array Explanation

- A 10 $\times$ 10 grid contains 100 icons, each representing one crop field.
- Color 22 icons (e.g., red) to show fields affected by pests.
- Leave the remaining 78 icons uncolored or neutral.

- This visual makes it immediately clear that only a small portion (22%) of fields are affected.



PROBLEM 38: Point Estimate Uncertainty

1. Confidence intervals

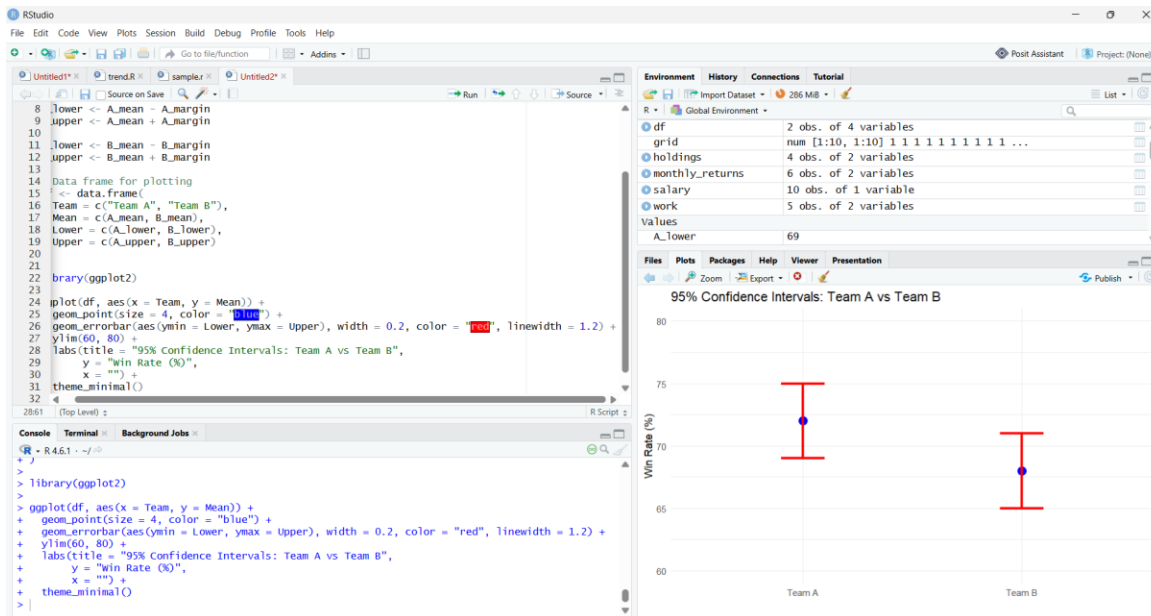
- Team A: $72\% \pm 3\% \rightarrow 95\% \text{ CI} = 69\% \text{ to } 75\%$
- Team B: $68\% \pm 3\% \rightarrow 95\% \text{ CI} = 65\% \text{ to } 71\%$

2. Do the intervals overlap?

- Overlap region: from 69% (max of lower bounds) to 71% (min of upper bounds).
- So, yes, the confidence intervals overlap.

3. Is Team A's higher figure a statistically meaningful lead?

Because the 95% confidence intervals overlap, we cannot claim a clear statistically significant lead for Team A based on these intervals alone. The observed difference (4 percentage points: 72% vs 68%) is not clearly beyond the uncertainty captured by the $\pm 3\%$ margins, so the higher win rate is not statistically meaningful at the 95% level from this CI comparison.



PROBLEM 39: Trend Uncertainty

1. Fit the linear trend

Using a regression of subscribers on time:

$$\hat{y} = 82.8 + 4.4x$$

This shows a steady increase of 4.4 units per period.

2. Compute residuals

Residuals = actual – predicted. They are small (around ± 2), meaning the linear model fits well.

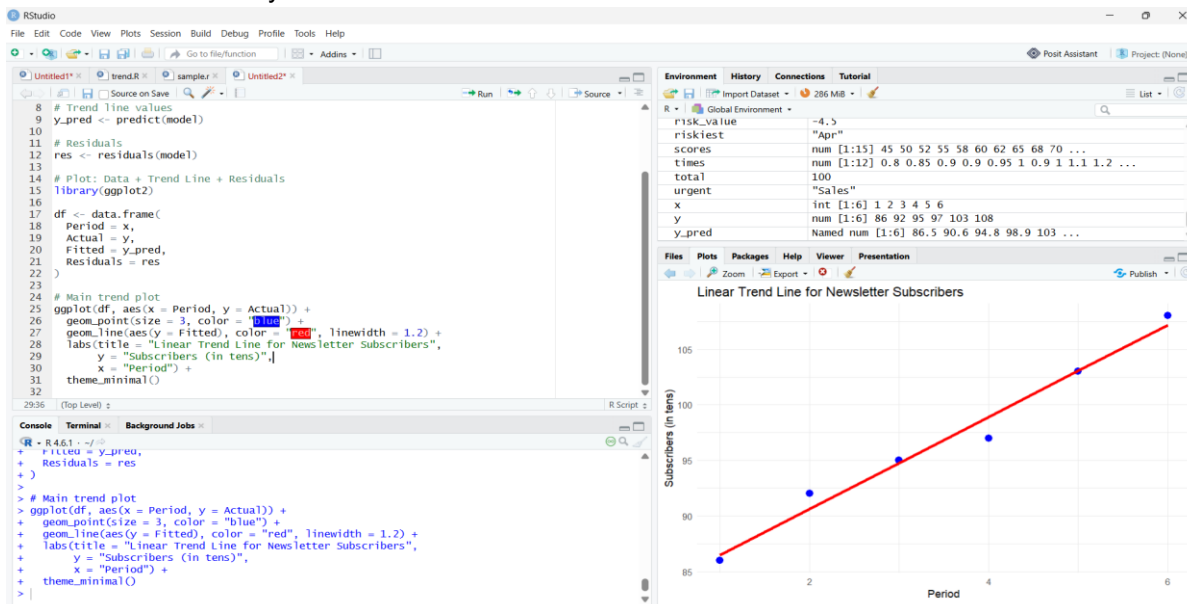
3. Standard deviation of residuals

Residual SD ≈ 2.0 This measures typical deviation from the trend line.

4. 95% uncertainty band

$$\text{Band width} = 1.96 \times \text{SD} \approx 3.9$$

So the 95% uncertainty band around the trend is ± 3.9 units.



PROBLEM 40: Hypothetical Outcome Plots (HOPs)

1. Hypothetical Outcome Plot frames

Each frame shows one possible daily outcome drawn from a distribution with:

- Mean: 300 loaves
- Standard deviation: 15 loaves

So every frame is a realistic daily production number around 300, sometimes a bit lower, sometimes a bit higher.

2. What a viewer learns from 5 frames vs a single error bar

- Single error bar (e.g., 300 ± 15 or a 95% CI):
 - Shows average and range of uncertainty in a static way.
 - Harder to feel how outcomes actually vary from day to day.
- Watching 5 frames in sequence (Hypothetical Outcome Plot):
 - Viewer sees day-to-day fluctuation around 300.
 - Understands that some days are below 300, some above, all still plausible.
 - Builds an intuitive sense of variability, not just a fixed interval.

